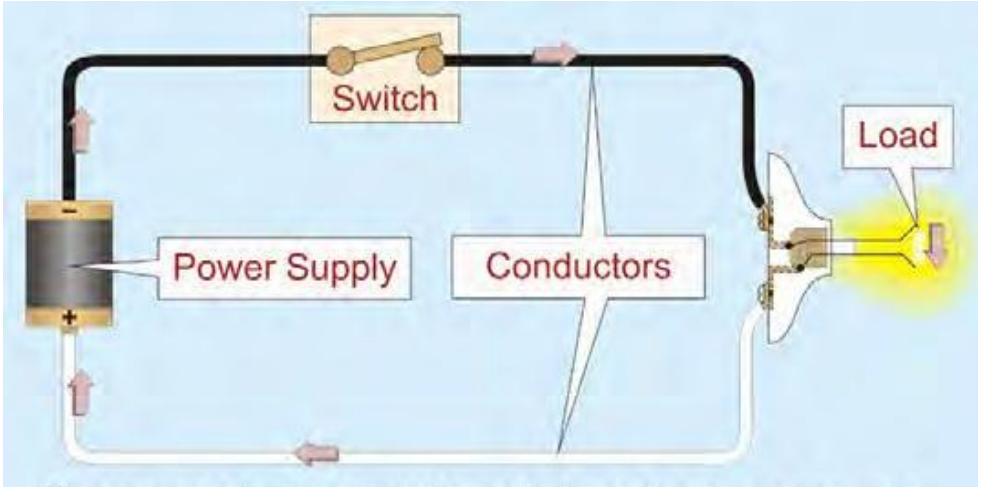
Fundamentals of Electronic Circuits

Electronics is the study and use of electrical components to control the flow of electricity and perform useful tasks, such as lighting a bulb, powering a fan, or operating electronic devices. At the heart of electronics are **circuits**, which are pathways for electric current to flow. Understanding circuits is the first step toward learning electronics.

Components of an Electronic Circuit

An electronic circuit is made up of several key components:

- **Power Source**: Provides energy to the circuit, usually a battery or DC power supply. In electronics, this is often low voltage (e.g., 1.5V to 12V) for safety and device compatibility.
- **Conductor**: Wires or traces on a circuit board that carry electric current from the source to other components.
- **Load**: Any component that uses electricity to perform work, such as an LED, motor, or buzzer. In electronics, the load often affects how the circuit behaves.
- **Switch**: Controls the flow of current by opening or closing the circuit. Switches can be simple buttons or more complex electronic switches like transistors.



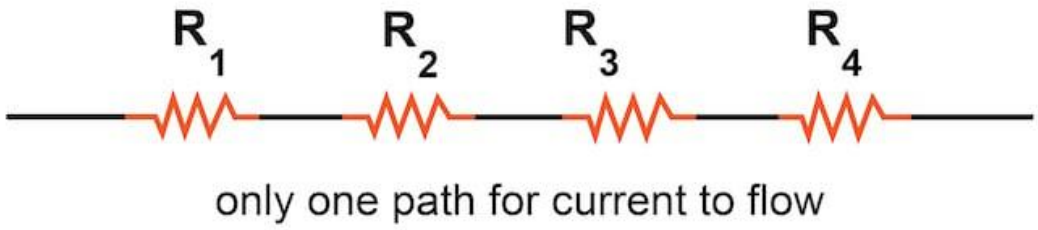
There must be a complete path from the power source, through the load, and back to the power source for electrons to flow.

Series and Parallel Circuits

Circuit components can be connected in two main ways: series and parallel. These basic connection methods can also be combined to form more complex series-parallel circuits.

Series Circuits

- Components are connected in a single path.
- Current is the same through all components.
- Voltage is divided across components.
- If one component fails, the entire circuit stops working.



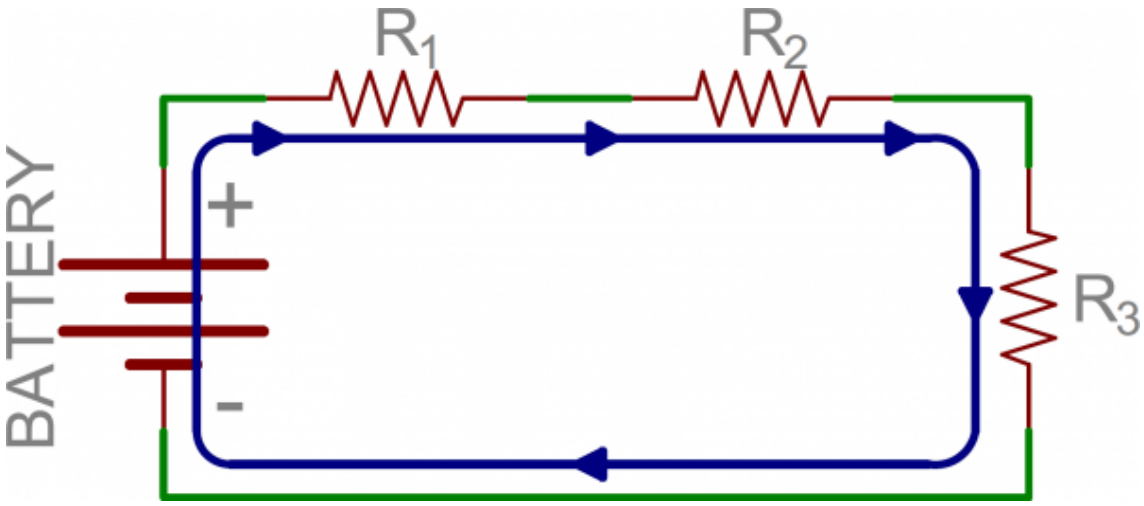
Formula: Rtotal = R1+R2+R3+…

Example: 2Ω + 3Ω + 5Ω = 10Ω

In a **series circuit**, all components are connected one after another, forming a single path for the electric current. This means that the same current flows through all components, but the voltage from the power source is divided among them depending on their resistance. The total resistance in a series circuit is equal to the sum of the resistances of each component.

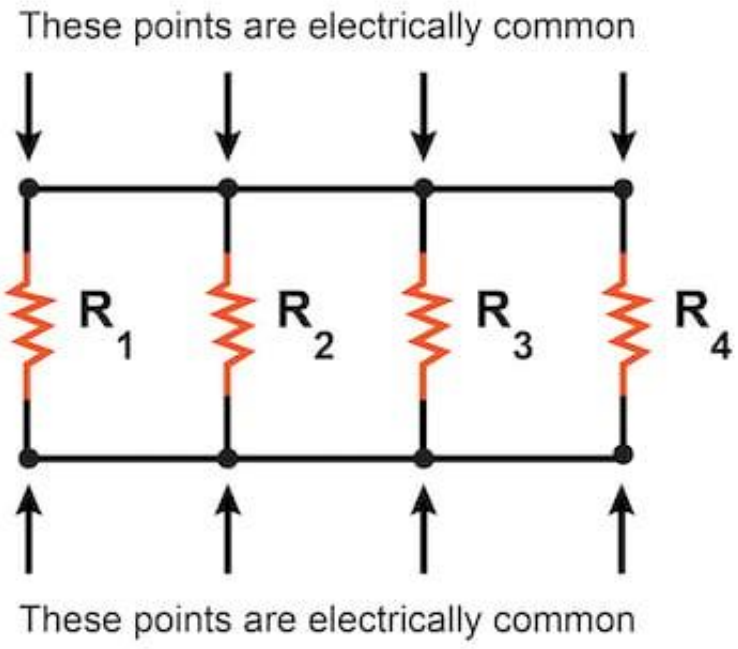
For example, if three resistors of 2Ω, 3Ω, and 5Ω are connected in series, the total resistance is 10Ω. If a 10V battery is applied, the current flowing through the circuit would be 1A. One important characteristic of

a series circuit is that if any component fails or is disconnected, the entire circuit stops working, because there is only a single path for current to flow.



Parallel Circuits

- Components are connected across multiple paths.
- Voltage is the same across each component.
- Current splits between the different paths.
- One component failing does not stop the others from working.

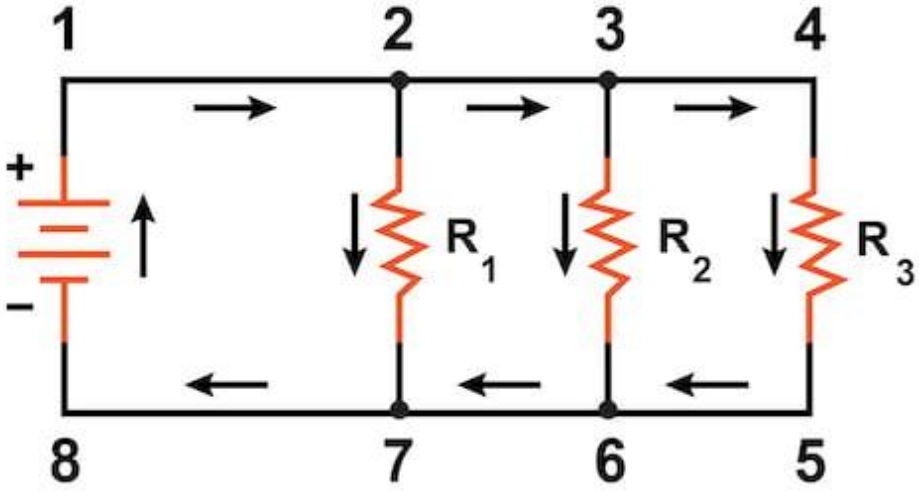


Formula: 1/Rtotal = 1/R1 + 1/R2 + 1/R3 +⋯+ 1/Rn

Example: 1/Rtotal = 1/2Ω + 1/3Ω + 1/6Ω = 0.5Ω + 0.33Ω + 0.16Ω = 1/1 = 1Ω

On the other hand, a **parallel circuit** has multiple paths for the current to flow because each component is connected across common points. This allows each component to receive the full voltage of the power source, while the total current is divided among the branches.

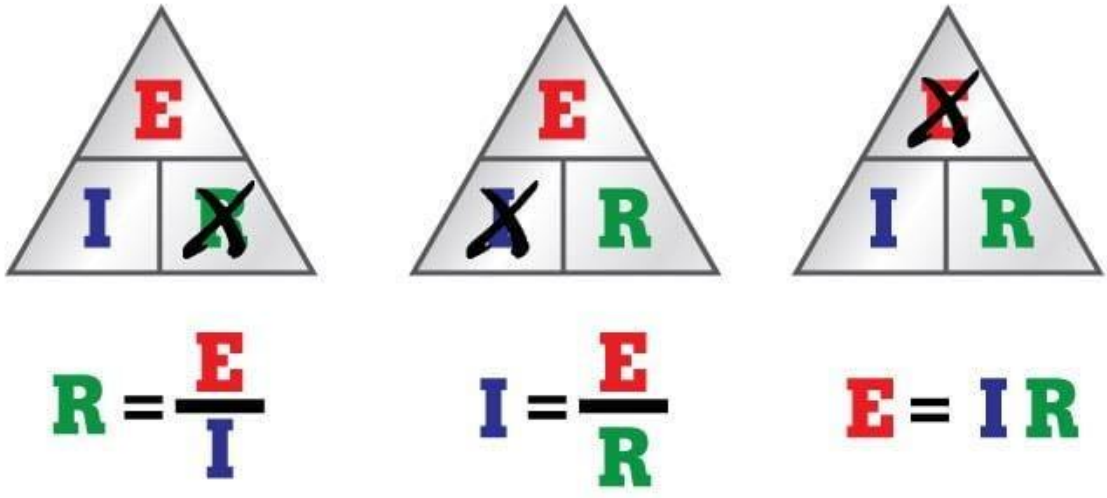
In electronics, parallel circuits are common because devices like LEDs or sensors often need the same voltage to operate properly.



Ohm's Law

Ohm's Law is the foundation of electronic circuit calculations. It shows the relationship between **voltage (V) or (E)**, **current (I)**, and **resistance (R)**:

It is sometimes easier to remember this Ohms law relationship by using pictures. Here the three quantities of V, I and R have been superimposed into a triangle (affectionately called the **Ohms Law Triangle**) giving voltage at the top with current and resistance below. This arrangement represents the actual position of each quantity within the Ohms law formulas.

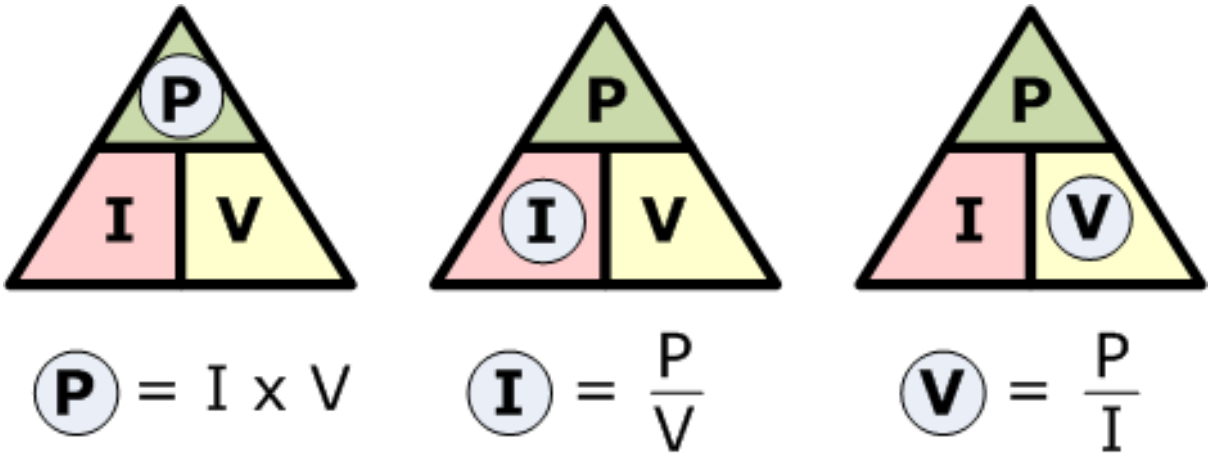


- **Voltage (V)**: Pushes electrons through the circuit (measured in **volts (V)**).
- **Current (I)**: Flow of electrons (measured in **amperes (A)**).
- **Resistance (R)**: Limits the flow of current (measured in **ohms (Ω)**).

In electronics, components like **resistors** are used to control the current flowing through sensitive devices like LEDs, preventing damage.

Power

In an electrical circuit, **Power (P)** is the rate at which electrical energy is used or consumed by a component. It tells us how much work a device can do using the current and voltage supplied. Power is measured in **watts (W)**.



But what is the formula for power. By using the Ohm's law equation and substituting for the values of V, I and R the electrical power formula found as seen above.

Ohms Law Matrix Table

Ohms Law Formulas				
Known Values	Resistance (R)	Current (I)	Voltage (V)	Power (P)
Current & Resistance	---	---	$V = IxR$	$P = I^2xR$
Voltage & Current	$R = \frac{V}{I}$	---	---	$P = VxI$
Power & Current	$R = \frac{P}{I^2}$	---	$V = \frac{P}{I}$	---
Voltage & Resistance	---	$I = \frac{V}{R}$	---	$P = \frac{V^2}{R}$
Power & Resistance	---	$I = \sqrt{\frac{P}{R}}$	$V = \sqrt{PxR}$	---
Voltage & Power	$R = \frac{V^2}{P}$	$I = \frac{P}{V}$	---	---